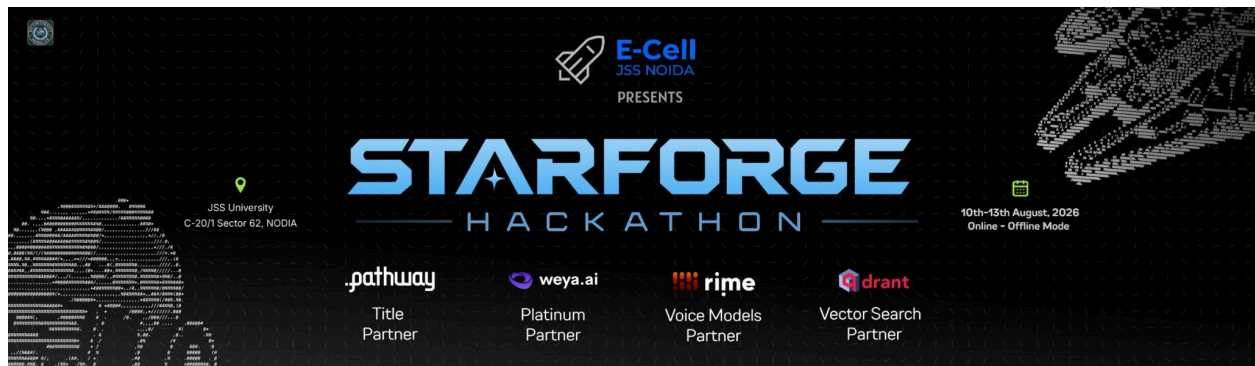




STARFORGE 2026 — HACKATHON

PROBLEM STATEMENT

Build the Present. Explore the Frontier.

StarForge 2026 gives you two ways to build at the frontier of AI.

Choose **one of two tracks**:

Before You Choose a Track

Still deciding between VoxForge and DragonForge? Use this side-by-side to see which track matches what you actually want to spend the next few days doing.

Choose VOXFORGE if you want to	Choose DRAGONFORGE if you want to
Build a real AI application	Explore AI research
Create a voice assistant or agent	Build tools for researchers
Focus on product development	Focus on evaluation & experimentation
Work with users and UX	Work with AI architecture and benchmarks

Choose VOXFORGE if you want to	Choose DRAGONFORGE if you want to
Primary Challenge: Building cohesive, low-latency, and reliable voice agents	Primary Challenge: Unpacking model reasoning, visualization, and explainability

Choose Your Track

VOXFORGE	DRAGONFORGE
Product engineering, Voice & agents, Retrieval, UX / workflows	Deep learning, Model internals, Experiments, Visualization
Build a working voice product. Rime speech must drive the core experience — not just decoration. Pair with Qdrant for smart retrieval. Must handle a real user problem end-to-end.	Research, explain, or evaluate Pathway's BDH post-transformer model. No need to train large models. Instead: build explainers, benchmarks, visualizations, or reproducibility studies.

Track 1 — VoxForge

Build a useful voice experience or AI voice agent.

Create a working product where **voice is essential to the experience** — not simply a chatbot with a microphone.

Your solution should use:

- **Rime** for voice generation
- **Qdrant** for meaningful retrieval, memory, routing, or evaluation
- Any suitable AI/LLM, speech recognition, orchestration, frontend, backend, or tool stack

Your voice agent should solve a real problem and demonstrate at least one challenging production problem such as:

- Real-time conversations and interruptions
- Low-latency voice interaction

- Multilingual voice experiences
- High-trust workflows
- Voice-based learning or accessibility
- Tool calling and task completion
- Memory and continuity across conversations
- Voice evaluation and testing

Eight possible VoxForge routes

Route	Focus
Real-time voice agents	Coherent turns, interruptions, recovery, reconnects, and full user-perceived latency.
Voice for high-trust workflows	Healthcare, insurance, finance, operations, public services, or another consequential task with confirmations, recovery, and human escalation.
Tools for voice builders	Compare prompts/voices, test pronunciation, catch regressions, measure latency, or evaluate what users hear.
Multilingual voice experiences	Test languages, names, numbers, domain vocabulary, and code switching; route voices deliberately.
Voice beyond the screen	Hands-free experiences for accessibility, mobility, hardware, laboratories, field work, or screen-impractical settings.
Learning out loud	Tutor, coach, interview partner, or practice tool using short spoken turns and useful feedback.
Voices with character	Interactive story, game, media experience, or character with a consistent voice and delivery.
Wildcard	A different voice-first idea, provided Rime solves a clear interaction problem and Qdrant makes a meaningful contribution.

Important

Voice must add real value.

If the same product works equally well as a text-based application, the idea does not sufficiently use the voice medium.

Qdrant must also have a **meaningful role**. Simply adding a vector database without affecting the product is not enough.

Track 2 — DragonForge

Explore the future of AI reasoning with Dragon Hatchling (BDH).

This track focuses on **research, experimentation, evaluation, visualization, and tooling** around Pathway's Dragon Hatchling (BDH) architecture.

You do **not** need to train a large AI model.

Instead, build something that helps people:

- Understand BDH better
- Evaluate reasoning capabilities
- Visualize model behavior
- Reproduce or improve existing experiments
- Study continual learning or memory
- Investigate long-horizon reasoning
- Explore interpretability
- Build evaluation or research tools

You can work directly with the public BDH implementation, existing models, benchmarks, released results, or previous community projects.

DragonForge Implementation Tracks

Work directly with public BDH	Work one level above the model
• Instrument bdh.py — inspect activations, RoPE, sparsity, attention • Build an explainer (Manim, web, notebook) mapping a BDH mechanism to the paper • Run small feasible experiments with the public implementation • Audit or reproduce an earlier community BDH project	• Build benchmark tools around long-horizon reasoning, memory, or interpretability • Use models available today for comparisons, failure analysis, or diagnostic views • Explain established BDH results (monosemantic synapses, sparse activity) with better pedagogy • Design so a larger BDH model could later enter the same evaluation framework

Example Ideas

1. BDH Visualization Create an interactive visualization explaining concepts such as attention, sparsity, RoPE, gating, or repeated levels.

- 2. Benchmark & Evaluation Tool** Build a framework to evaluate long-horizon reasoning, memory, or continual learning.
 - 3. BDH Microscope** Instrument the public implementation to visualize activations, sparsity, attention, or tensor behavior.
 - 4. Research Reproduction** Reproduce an existing BDH experiment and improve its measurements, testing, or explanation.
 - 5. Interpretability Tool** Create a tool for exploring concept-selective or monosemantic behavior.
 - 6. Research Question Explorer** Study an important BDH research direction and build an experiment or evaluation framework that could later be used with larger BDH models .
-

Mandatory Submission Requirements

Every team must submit a **public GitHub repository**.

Your repository should contain:

1. Problem / Research Question

Clearly explain what you are solving or investigating.

2. Solution

Explain how your project works.

3. Architecture / Evaluation Flow

Include a simple architecture diagram or experiment flow.

4. Working Proof

Show at least one:

- Live working feature
- Reproducible experiment
- Test
- Benchmark result
- Visualization
- Measurable improvement

5. README

Include clear setup and run instructions.

7. Limitations

Clearly mention what your project does and does not prove.

8. Team Contributions

Mention what each team member contributed.

These requirements are based on the official StarForge submission structure.

Important Rules

- A public GitHub repository is mandatory.
 - AI-assisted coding is allowed.
 - Teams must be able to explain their implementation.
 - Do not expose API keys or credentials.
 - Use synthetic or de-identified data for sensitive domains.
 - Verify that your submitted project actually works.
 - Do not submit only a concept, presentation, or static prototype.
 - For VoxForge, Rime must be essential to the main voice experience.
 - For VoxForge, Qdrant must have a meaningful functional role.
 - For DragonForge, clearly distinguish between established findings, your own measurements, and exploratory ideas.
-

Judges will look for:

VoxForge

- **Problem & usefulness — 30%**
- **Voice experience — 25%**
- **Use of Rime — 20%**
- **Technical execution — 15%**
- **Demo — 10%**

DragonForge

- **Builder contribution — 30%**

- Technical & research fidelity — 25%
 - Evidence & evaluation — 20%
 - BDH relevance & next question — 15%
 - GitHub & communication — 10%
-

Choose Your Mission

VOXFORGE

Build a voice experience people would genuinely prefer to use.

DRAGONFORGE

Build something that helps us understand, evaluate, or explore the future of AI reasoning.

Build the present. Explore the frontier. Leave a signal worth following.

Reference Links & Learning Resources

Participants can use the following resources to understand the technologies, tools, and research concepts involved in StarForge 2026.

VoxForge — Voice AI

Rime

- Rime Documentation: <https://docs.rime.ai/docs/introduction>
- Rime Quickstart: <https://docs.rime.ai/docs/quickstart-cli>
- Rime Models: <https://docs.rime.ai/docs/models>
- Rime Voices: <https://docs.rime.ai/docs/voices>
- Rime Prompting Guide: <https://docs.rime.ai/docs/prompting>

Rime's documentation provides information about Coda, Mist v3, voices, model selection, prompting, and getting started with voice generation.

Qdrant

- **Qdrant Documentation:** <https://qdrant.tech/documentation/>
- **Qdrant Search:** <https://qdrant.tech/documentation/search/>
- **Qdrant Hybrid Search:**
<https://qdrant.tech/documentation/search/text-search/hybrid-search/>
- **Qdrant Filtering:** <https://qdrant.tech/documentation/search/filtering/>
- **Qdrant Hybrid Search Tutorial:**
<https://qdrant.tech/documentation/tutorials-and-examples/cloud-inference-hybrid-search/>

These resources cover vector search, retrieval, metadata filtering, hybrid search, and other capabilities that can be used to give Qdrant a meaningful role in your project.

Weya AI

- **Weya AI:** <https://www.weya.ai/>
- **Weya AI Voice:** <https://www.weya.ai/product/voice-ai>

Use these resources to understand production voice-agent workflows, customer calls, multilingual agents, memory, human handoff, and workflow automation.

DragonForge — BDH & AI Research

Pathway & Dragon Hatchling (BDH)

- **Pathway:** <https://pathway.com/>
- **BDH Research:** <https://pathway.com/>
- **BDH Explainer:** <https://pathway.com/research/bdh-explainer>
- **BDH GitHub Repository:** <https://github.com/pathwaycom/bdh>
- **Dragon Hatchling Paper:** <https://arxiv.org/abs/2509.26507>
- **Pathway BDH Paper Page:**
<https://pathway.com/news/the-dragon-hatchling-the-missing-link-between-the-transformer-and-models-of-the-brain>

The BDH resources introduce the Dragon Hatchling architecture and provide the public implementation and research paper that participants can use as a foundation.

Interactive AI / Transformer References

- **Transformer Explainer — Interactive Demo:** <https://poloclub.github.io/transformer-explainer/>
- **Transformer Explainer — GitHub:** <https://github.com/poloclub/transformer-explainer>

These can be useful for participants interested in building interactive explainers and visualizations for AI architectures.

Community Projects & Inspiration

Participants may also inspect previous BDH-related projects mentioned in the official problem statement.

These projects are provided **for learning and inspiration only**. Teams should build their own original solutions and aim to improve upon existing work through better experiments, measurements, testing, reproducibility, or explanations.

- **Model Merging + Visual Explainer:** <https://github.com/Kumarutkarsh94/70>
- **Game of Life: Interpretable Circuits:** https://github.com/Choco-princess/BDH_Nexus
- **Legal Violation Classifier:** <https://github.com/Deb044/LegalClassifier>
- **BDH Attention in a Diffusion Model:** <https://github.com/spandan11106/BDH-Explainer>

These community projects are specifically referenced in the official StarForge problem statement as examples to inspect, reproduce, improve, or learn from.

Important

Reference links are provided to help participants **learn the technologies and understand the problem space**.

Participants are expected to:

- Build an original solution.
 - Properly credit external code, datasets, models, and resources.
 - Clearly mention the use of AI-assisted coding or external tools where applicable.
 - Not expose API keys or other credentials.
 - Follow the submission and security requirements of the hackathon.
-

What We Expect in Round 1

Round 1 is evaluated on a PPT and video, not a live build — here's exactly what to prepare and submit.

- PPT (Official Template)
- 2-minute explanation video
- Public GitHub repository (if available)
- Working prototype/demo (recommended)
- Google Drive link to the video
- Team details

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